

Small Bank Lending in the Era of Fintech and Shadow Banks: A Sideshow?

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Amid the emerging dominance of nonbanks, small banks use key financing advantages to persist in the mortgage market. We provide evidence of the heterogeneous impact of two shocks to the supply of mortgage credit: postcrisis regulatory burden and GSE financing cost changes. Small banks exploit regulation disproportionately affecting the largest four banks (Big4) and their ability to lend on balance sheet to strongly substitute for the retreating Big4. The erasure of guarantee fee (g-fee) discounts for large lenders facilitates small bank growth in GSE lending. Small banks also grow balance sheet loans in areas more exposed to g-fee hikes. (*JEL* G2, R31, L1)

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The U.S. mortgage market has undergone massive shifts since the 2008 financial crisis. Nonbank lenders have taken advantage of heightened postcrisis bank regulation and technological advancements to surpass traditional banks in mortgage lending (Buchak et al. 2018; Fuster et al. 2019). These headwinds for traditional banks would seem particularly challenging for smaller banks, which lack the economies of scale of larger banks and the technological firepower of fintech lenders. Do small, local banks still have a role to play in this new era of mortgage lending? The answer to this question has implications for a wide range of issues, including housing rents, wealth inequality, systemic risk, and financial regulation.¹ In this paper, we uncover compelling new facts about the

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¹ Gete and Reher (2018) show that banking retreats lead to higher housing rents and mortgage denial rates, and D'Acunto and Rossi (2022) show that the areas with a lower share of big banks experience a redistribution

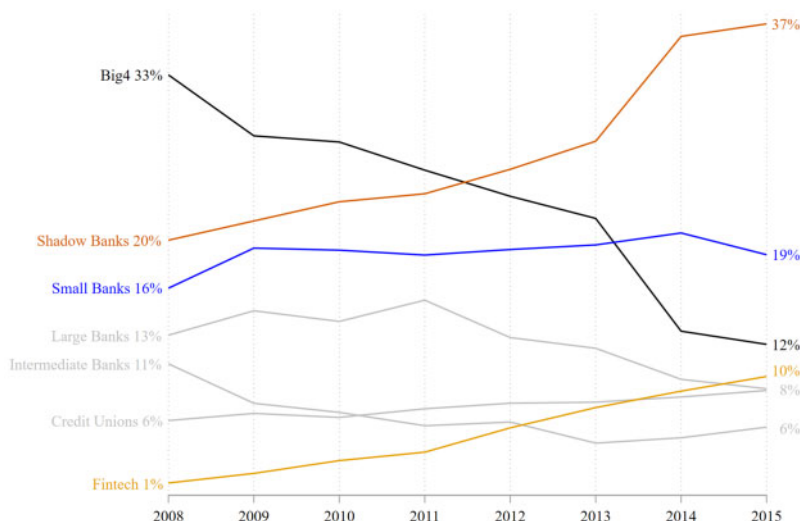


Figure 1
Aggregate lending shares over time

This figure presents the aggregate mortgage origination market shares for each lending class over time. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018).

enduring importance of small banks and show the channels through which they persevere.

Figure 1 presents a stark new fact: despite increased regulation, compliance costs, and technological disruption from competitors, small banks' (assets less than \$10 billion) position in the mortgage origination market has been remarkably stable with their share growing from 16% in 2008 to 19% in 2015. They have maintained this market share despite a surge from nonbank competitors, who have increased from 21% to 47%. Small banks' persistence stands in sharp contrast with all other traditional banks, whose collective market share dropped by more than half. The market share of the "Big Four" banks (Bank of America, Citibank, JP Morgan, Wells Fargo) dropped from 33% to 12%, "large" banks (assets between \$1tr and \$50bn) dropped from 13% to 8% and those of "intermediate" banks (assets between \$50bn and \$10bn) dropped from 11% to 6% from 2008 to 2015. This new fact about heterogeneity *within*

of credit supply away from smaller loans toward larger (jumbo) loans. Kim et al. (2018) raise concerns about systemic risk because of the inherent vulnerability of nonbanks to liquidity pressures. Buchak et al. (2020) and Elliott, Meisenzahl, and Peydró (2019) have shown the implications of composition on the efficacy of capital regulation and monetary policy transmission.

traditional bank lenders is masked when we restrict the analysis to the traditional bank versus nonbank margin.

How have small banks remained relevant during a time when all other classes of traditional banks significantly declined? To answer this question, we show the heterogeneous impact of two major shocks to the supply side of the mortgage market that each provide an advantage for small banks: differential postcrisis bank regulation and changes to financing costs of loans sold to the government-sponsored enterprises (GSEs), Fannie Mae and Freddie Mac. For the first shock, we provide evidence that the massive increase in regulatory burden following the crisis fell disproportionately on the Big4 banks, and the Big4 had negative subsequent growth in virtually all counties. We find that small banks grow most in areas previously dominated by the retreating Big4, and their local response to Big4 withdrawal is greater than that of any other lender class, including nonbanks. We find strong small bank substitution in the segment of loans held on balance sheet, an area of comparative advantage over nonbanks (Buchak et al. 2020). For the second shock, we show large changes to the cost of financing loans through the GSEs provided small banks with growth opportunities in both GSE and balance sheet lending. Thus, against a broader tide of nonbank lending growth across the country, small banks have exploited balance sheet lending and financing cost advantages to remain important players in the mortgage market.

Following the financial crisis, traditional banks faced numerous regulatory shocks, including increased capital requirements, significant changes in the regulatory treatment of mortgage servicing rights, and sizable lawsuits relating to precrisis activities (Buchak et al. 2018). Traditional bank lenders range from very small community banks to the largest banks exceeding one trillion dollars in assets. Uneven contours in regulation across bank sizes (Bouwman and Johnson 2018) may yield opportunity for regulatory arbitrage within traditional banking, similar to the forces that have facilitated broad nonbank growth relative to banks as a whole. We find that postcrisis regulatory burden fell particularly hard on the Big4. The Big4 faced more than \$100 billion in fines related to their mortgage business, and that sum is five times more than all other traditional banks combined. Also, they faced the steepest demands to increase their equity capital; were disproportionately negatively affected by the change in capital rules regarding the treatment of mortgage servicing rights; and were the only banks initially designated as global systemically important banks. Figure 2 provides new direct evidence that the incidence of key postcrisis regulatory shocks disproportionately fell on the Big4.

In light of the evidence of disproportionate change in regulation, we examine county-level substitution patterns across different sizes of banks (Big4, large, intermediate, and small banks) as well as other lender classes (credit unions, nonbanks) by correlating loan growth rates from 2008 to 2015. The patterns of substitution with the Big4, whose loan growth was weakly negative for all counties in our sample, are striking. While each lender class's county-level

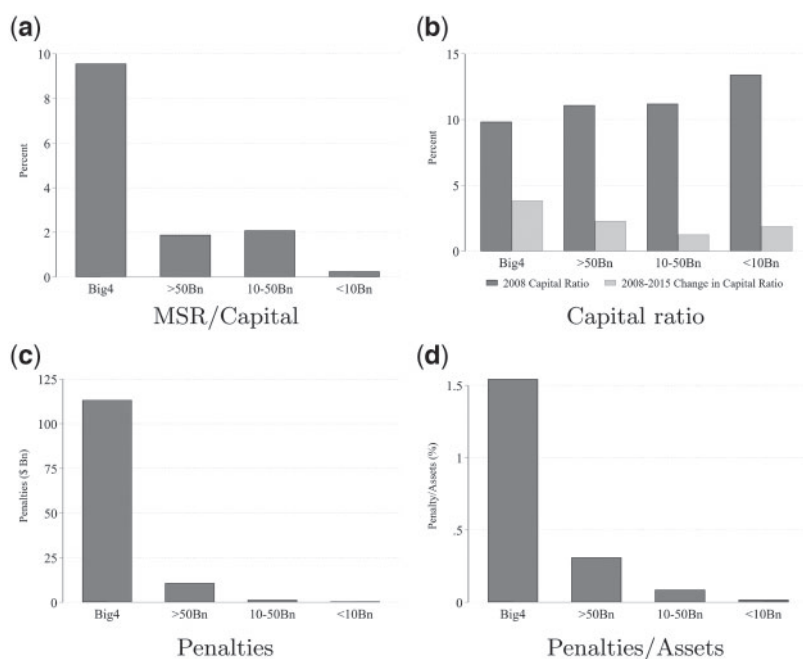


Figure 2
The Incidence of Regulatory Shocks Across Banks

This figure presents the degree to which banks of different sizes were exposed to postcrisis regulatory shocks (described in detail in Section 3). Panel a plots the mortgage servicing rights as a share of regulatory capital; panel b presents the 2008 tier1 capital and the changes from 2008 to 2015; panel c presents the raw magnitude of mortgage-related fines; and panel d presents those fines as a share of total assets. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; and *Small banks* includes banks with assets less than \$10 billion.

growth is negatively correlated with Big4 growth, this negative correlation is largest for small banks. Large and intermediate banks—whose increase in regulatory burden was between that of the Big4 and small banks—occupy somewhat of a middle ground in the sense that they only show a modest negative correlation with the Big4 and small banks.

To study relative elasticities of credit supply to the Big4 withdrawal, we regress the 2008–2015 (county-level) lending growth for each lender-class on the growth in Big4 lending during that period.² A 10-percentage-point (pp) larger decrease in Big4 growth corresponds to an increase in small-bank lending growth of 6.6 pp. There is no economically meaningful response from large or intermediate banks to the Big4 retreat. For nonbanks, a 10-pp lower Big4 growth corresponds to 1.3- and 0.5-pp higher growth from shadow banks and

² While we focus on the 2008–2015 period to be consistent with prior work (Buchak et al. 2018), our results are robust to adjusting these start and end dates.

fintech lenders. While these nonbanks have substantial growth across the nation, they are much less sensitive to the local variations in the Big4 retreat than are small banks.

To interpret our estimates as the relative degree of substitution on the supply side, an ideal experiment would randomly assign Big4 growth rates across counties and observe other lenders' responses. In the absence of the ideal, demand-side factors could cloud our interpretation. For example, consumer demand for Big4 products might have systematically declined at a time when demand for small banks' products simultaneously increased. We directly control for observable factors that could relate to demand shocks by including changes in house prices; county demographics, such as population, minority share, and per capita income; and the local lending market's competitive structure in our regressions. Our within-county lending growth estimates also account for unobserved time-invariant county-specific factors.

To further address potential concerns that time-varying demand-side factors explain our results, we use two complementary instrumental variables (IV) strategies. Our first IV captures the combined effect of multiple coincident regulatory shocks in reduced-form by using the 2008 county mortgage origination share of the Big4 as an instrument for the subsequent 2008–2015 growth in Big4 lending.³ In 2008, crisis-related fines had yet to be levied, and the Dodd-Frank Act with its increases in the regulatory burden had not yet passed. Using predetermined levels to instrument for changes helps us isolate the variation in the credit supply of the Big4 that is unlikely to be related to future demand shocks. Instead, the variation is driven by the differing scope to reduce mortgage lending, which is likely higher in counties in which the Big4 had a larger initial mortgage share. Our second IV uses directly the differential incidence in postcrisis lawsuit exposure to instrument for Big4 growth rates. Since lawsuits primarily targeted the Big4 banks, we expect areas more exposed to this regulatory shock to experience greater Big4 withdrawal. We construct a 2008 origination-weighted measure of lawsuit exposure as the instrument for 2008–2015 growth in Big4 lending. Both instruments have strong first stages and yield similar point estimates: a 10-pp larger decline in Big4 growth corresponds to a 9- to 10-pp increase in small bank growth.

Our ordinary least squares (OLS) and instrumental variables (IV) results show the local response of nonbanks to the Big4 growth to be much less sensitive than that of small banks. Since nonbanks hold a significant advantage over all traditional banks due to lower regulatory burden (Buchak et al. 2018), how is it that small banks substitute so strongly for the Big4? While nonbanks securitize the vast majority of the loans they originate, traditional banks maintain a distinct advantage in their ability to originate loans that remain

³ The use of a predetermined measure of the distribution of Big4 shares is similar to the IV strategies in D'Acunto and Rossi (2022) and Gete and Reher (2018), who seek to avoid potentially confounding demand shocks resulting from financial crisis events.

on-balance-sheet. We find that the small bank substitution for the Big4 is twice as large for balance sheet lending compared to their sensitivity to non-balance-sheet lending. The especially higher substitution in balance sheet loans is consistent with small banks using a key source of competitive advantage over nonbanks (Buchak et al. 2020). This channel is not obvious because banks can adjust on the balance sheet retention margin—facing a marginal increase in regulatory burden, banks can reduce loans retained on their balance in favor of an originate-to-distribute (OTD) model— or they can reduce their lending altogether. When we examine growth in the two modes of financing (balance sheet lending vs. OTD), we find overall balance sheet lending for all traditional banks (including small banks) to be negative on average, which is consistent with loan retention becoming costlier due to capital pressures. In the GSE segment, however, small bank average growth is greater than that of all other traditional banks. To understand this striking growth pattern, we study a second major shock to the supply side: significant and heterogeneous changes in the cost of financing loans through the GSEs, Fannie Mae and Freddie Mac.

Lenders that sell to the GSEs pay guarantee fees (g-fees) to account for the default risk of the loans they sell them. These g-fees allow the GSEs to guarantee principal and interest payments (i.e., eliminate default risk) to investors in their mortgage-backed securities. Following mandates from the Federal Housing Finance Agency (FHFA) beginning in 2011, the g-fees charged by GSEs underwent major changes. During 2011–2015, (1) a nearly 10-basis-points (bps) discount for the top-volume GSE lenders was eliminated, effectively leveling the funding costs across all lenders for loans sold to the GSEs (change to *parity*), and (2) the average level of g-fees approximately doubled from about 30 to 60 bps (change in *level*). The nature of these changes provides specific testable predictions to investigate whether these shocks to costs play an important role in explaining small banks' lending patterns.

Consistent with the parity in g-fees being a new source of opportunity for small banks, we find strong, widespread growth in GSE lending by small banks. Their growth in this segment is largest in counties previously dominated by large GSE lenders (whose g-fee advantage was erased by parity) before the change. While the Big4 were among the top lenders, this channel is not just a consequence of disproportionate regulation on the largest banks. We find similarly strong growth of small bank GSE lending in areas dominated by top GSE lenders that are *nonbanks*, establishing the distinct role of the g-fee financing cost shocks.

The increase in the level of g-fees across all lenders also benefits small banks through a different market segment. All else equal, the increase in GSE financing costs would render balance sheet financing of loans more attractive at the margin, even for loans that meet GSE standards (Hurst et al. 2016; Buchak et al. 2020). We find that small banks had substantial gains in balance sheet lending in areas that previously had higher GSE financing activity (i.e., greater exposure to the increase in levels of g-fees) before the fee hikes.

In sum, mortgage lending activity from small banks has remained remarkably stable following the financial crisis, even in the face of increasing regulatory burden and the rise in competition from nonbanks. We find that two significant shocks to the supply side of the mortgage market—differential bank regulation and shocks to GSE financing costs—along with small banks’ ability to lend on balance sheet play key roles in the continuing importance of small banks in this market.

1. Related Literature

Our paper is related to recent work on changes in the U.S. mortgage market, including Buchak et al. (2018, 2020) and Fuster et al. (2019). These papers study the rise of “shadow banks,” a term broadly referring to nonbank lenders, including fintech and nonfintech lenders.⁴ While traditional banks have declined in their share of residential lending upon facing greater regulatory scrutiny and capital costs, nonbanks take advantage of regulatory arbitrage to grow their lending. We complement this finding from Buchak et al. (2018) by showing substitution between the Big4 and small banks as a result of disproportionate regulatory burden on the very largest players. Our study is among the first to shed light on substitution patterns across banks of different sizes *within* traditional banking.

Buchak et al. (2020) study the choice of banks to securitize loans versus financing them on balance sheet. Traditional banks have a distinct advantage over nonbanks in balance sheet financing, so the expansion of nonbanks is limited by their inability to hold loans on their books. Indeed, we find that lending on balance sheet plays a key role in our results and are an important feature to understanding small banks endurance in the face of nonbank competition. Our work further complements Buchak et al. (2020) by studying a shock to GSE financing costs that relates to a key prediction of their mechanism. Specifically, we show that small banks increase balance sheet lending more in areas in which the increase in GSE financing costs are more prominent.

Our work relates to a large literature on the unique role of small, local banks. While sometimes viewed as less sophisticated and set back from the technological forefront, small banks have been critical to relationship lending (Bolton et al. 2016; Boot and Thakor 2000; Hein, Koch, and MacDonald 2005; Petersen and Rajan 1994) and cultivating the local knowledge about the borrowers they serve (Berger et al. 2005; Loutskina and Strahan 2011; Gilje 2019). We add to this line of work by highlighting small banks’ ability to retain loans on balance sheet as a source of advantage against technologically superior competitors.

⁴ Gorton et al. (2010), Adrian and Ashcraft (2012), and Acharya, Schnabl, and Suarez (2013), among others, focus on the role of regulatory arbitrage in the shadow banking sector during the lead-up to the crisis.

Our paper relates to Cortes et al. (2020), who study the substitution by local lenders in small business lending. In a market still dominated by banks, Cortes et al. (2020) show that where large banks retreat because of stress testing and higher regulatory costs, small banks increase their activity. However, the competitive and institutional dynamics (such as the pervasiveness of securitization) make the mortgage market very different from small business lending, so it is unclear whether their conclusions carry over to our paper. In mortgage lending, competition from nonbanks is significantly higher (over 47% market share in 2015), and these lenders face lower regulatory costs relative to small banks. If anything, nonbanks may be *better* positioned to fill the gap left by the largest banks.⁵ Further, technology plays a critical role in mortgage lending (Fuster et al. 2019), and local banks would seem to be at a substantial disadvantage relative to fintech lenders along a technological dimension.

We add to an emerging body of work on the consequences of heterogeneity in bank regulation. Behn, Haselmann, and Vig (2022) show that large banks benefited more from the introduction of model-based capital regulation than smaller banks. Bouwman and Johnson (2018) show that banks just below the asset size thresholds of Dodd-Frank alter their growth to avoid crossing these thresholds and bearing increased regulatory costs. Our paper shows how the contours of regulation generate within-traditional bank substitution in the supply of mortgages. Benetton (2021) finds that differential regulatory treatment for large U.K. lenders lowers their cost of mortgage originations. In counterfactual analyses, the paper shows that standardizing capital regulation across banks would increase costs for large lenders and reduce market concentration. Our paper describes a novel setting that yields results consistent with this prediction, where changes in GSE guarantee fees confer a cost advantage to small banks relative to their larger counterparts, resulting in strong growth by small lenders in this market.

Finally, our work relates to Gabaix (2011), who shows that the actions of a few large players in an industry can have substantial ripple effects throughout the market. While much of the literature on large bank behavior has focused on too-big-to-fail (TBTF) in terms of systemic risk and potentially inefficient risk-taking, among other issues, we show that the behavior of TBTF banks can have far-reaching implications even in normal times.

2. Data and Summary Statistics

Mortgage originations data are from the Loan Application Register (LAR) files of the Home Mortgage Disclosure Act (HMDA) database. In our analysis, we include all originated loans whose purpose was either home purchase or refinancing. We measure lending activity as the number of loans for our tests,

⁵ For example, De Roure, Pelizzon, and Thakor (2022) use German data on consumer credit to show that fintech (P2P) lending grows when banks are shocked with a higher regulatory burden.

although we could similarly analyze loan amounts. Our empirical analysis requires classification to the lender class level or individual lender level. HMDA lenders (“respondents”) only report direct parent affiliations instead of the ultimate holding company Avery, Brevoort, and Canner (2007). We use the HMDA Lender file from Robert Avery to associate depository institutions and mortgage companies with their ultimate parents. This avoids a fundamental problem of misclassifying subsidiaries: affiliated mortgage companies are folded into the holding organization and then reported on an aggregated basis in our sample. Nonbanks are nondepository institutions that participate in mortgage credit origination. We subdivide nonbanks into those that specialize in using financial technology (*fintech lenders*) and those that do not (*shadow bank lenders*). We rely on the Buchak et al. (2018) classification of fintech: lenders with a significant online presence, so that nearly all of the application process occurs online and without human interaction. We then aggregate the data at the lender-county level.

Where applicable, the Avery file also includes the RSSD ID for the lender, which allows us to match the bank HMDA lenders to their call report (bank-level) and their location and deposit data (bank-county level). We classify each lender into one of the following seven “lender classes”: Big4 (Bank of America, Citi, JP Morgan, Wells Fargo), large banks (assets between \$50 billion and \$1 trillion), intermediate banks (assets between \$10 billion and \$50 billion), small banks (assets less than \$10 billion), credit unions, shadow banks, and fintech lenders. Our size divisions are motivated by thresholds specified in the Dodd-Frank Act and are consistent with prior literature (Bouwman and Johnson 2018; Cortes et al. 2020).

Our main data set is a lender class $\times$ county panel from 2008 to 2015. Most tests include controls for county-level demographics and house prices, which we collect from the Census and FHFA, respectively. Our main variables of interest are county-level c lending growth for each lender class LC over a specified time period, either 2008–2015 or 2011–2015. Following Buchak et al. (2018), we compute this as follows:

$$Growth_c^{LC} = 100 \times \left[\frac{Originations_{c,2015}^{LC} - Originations_{c,2008}^{LC}}{Originations_{c,2008}^{All}} \right] \quad (1)$$

Column 1 of Table 1 shows average county-level lending growth from 2008 to 2015 for each lender class. The county-level averages broadly reflect the patterns shown in Figure 1. While all other traditional banks (Big4, large, and intermediate) have negative average growth, the average county-level small bank lending has increased 1.5 pp during our sample period, and credit unions also enjoyed modest growth of about 2.4 pp. The positive lending growth is notable given the substantial gains from nonbank competitors (9 pp for shadow banks and 7 pp for fintech) who have benefited from postcrisis bank regulation.

Some tests focus on loans retained on the balance sheets of lenders. The HMDA data record whether the loan was sold during the calendar when it was

Table 1
Summary statistics: County-level loan growth by lender class

	2008–2015			2011–2015		
	(1) All	(2) BalanceSheet	(3) GSE	(4) All	(5) BalanceSheet	(6) GSE
Big4	−19.96 (8.61)	−19.85 (12.41)	−30.79 (17.63)	−10.84 (6.97)	−8.40 (9.50)	−20.32 (15.17)
Large banks	−4.61 (6.17)	−4.71 (9.12)	−4.35 (12.37)	−6.60 (6.78)	−9.00 (13.31)	−7.57 (11.98)
Intermediate banks	−4.15 (6.19)	−2.48 (6.87)	−5.10 (12.13)	0.37 (5.02)	1.17 (7.52)	0.74 (9.06)
Small banks	1.46 (18.44)	−6.09 (26.97)	13.40 (23.29)	0.14 (17.28)	1.56 (30.42)	4.91 (22.71)
Credit unions	2.35 (5.11)	1.68 (9.13)	3.71 (8.14)	1.93 (5.54)	3.68 (11.96)	1.41 (7.99)
Shadow banks	9.28 (13.38)	−2.60 (7.63)	14.33 (21.51)	10.39 (12.56)	2.03 (7.47)	15.38 (21.14)
Fintech	6.93 (4.61)	1.38 (2.07)	17.20 (13.83)	5.66 (5.21)	1.31 (3.36)	14.47 (14.86)

Mean coefficients; SD are in parentheses.

This table presents county-level loan growth summary statistics for each lender class from 2008 to 2015 (expressed as percentage points). See Section 2 for the construction of this variable. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018). BalanceSheet loans are those classified in HMDA as “not sold” or “sold to an affiliate,” and GSE loans are loans sold to Fannie Mae or Freddie Mac.

originated, and we classify loans as held “on-balance-sheet” if HMDA classifies them as “not sold” or “sold to an affiliate.” This represents an upper bound on the loans retained on the balance sheet. Loans made in late December, for example, may not have time to be securitized in that calendar year.⁶ We classify loans as “GSE loans” if the HMDA data flag them as sold to Fannie Mae or Freddie Mac. Columns 2 and 3 of Table 1 shows county-level averages in balance sheet and GSE lending growth for each lender class from 2008 to 2015.⁷ Columns 4–6 specifically show the respective growth rates for 2011–2015, a time period that is relevant for our later tests examining the consequences of changes to the costs of selling loans to the GSEs.

2.1 Supply side shock measures

2.1.1 Regulatory shocks. Our measures of regulatory shocks broadly follow Buchak et al. (2018) (cf. Section 7 and Equations (10), (11), and (12)). We use regulatory capital measures and mortgage servicing rights from the call reports,

⁶ Seasonality in local real estate markets is unlikely to bias our results because our tests examine growth in lending within the same county over time.

⁷ We present the averages for balance sheet lending and GSE because they are most relevant for the shocks we examine. We do not separately consider other loan types, such as FHA loans, but those are captured in our broad measure of total lending.

and data on fines from the violation tracker database maintained by Good Jobs First.⁸

2.1.2 Shocks to GSE financing costs. To examine the extent that lenders are affected by changes to GSE guarantee fees (g-fee shocks), we compute each lender's "GSE Seller" rank in 2011 prior to the large shifts in g-fee costs schedules. Lenders with a greater volume of loans sold to GSEs enjoyed what amounts to volume discounts, and these were erased during 2011-2015. Thus, such lenders experienced a relatively larger impact of g-fee changes. Our empirical tests use a county-level measure of the impact of g-fee shocks, which we compute as follows for 2011. First, we calculate the total number of loans sold to Fannie Mae and Freddie Mac (which includes origination and loans purchased from other lenders to be resold to the GSEs). Then we rank the top-10 sellers based on the total GSE loans sold. While the "top-10" cutoff is somewhat arbitrary, our results are not sensitive to this particular definition. Rather, we find consistent results using the top-50 sellers. Finally, for each county, we compute the share of loan origination by those top lenders:⁹

$$Top10GSELenderShare_{c,2011} = \frac{Originations_{l,c,2011}^{Top10GSELenders}}{Originations_{c,2011}^{All}} \quad (2)$$

3. The Differential Incidence of Regulatory Shocks across Banks

The passage of the Dodd-Frank Act was a landmark change in bank regulation following the financial crisis. In addition, there were sizable fines levied on several lenders for mortgage-related activities during the run-up to the crisis. In this section, we provide evidence on the differential incidence of regulatory shocks across banks of different sizes with the size thresholds based on the broad contours of the regulatory changes (Cortes et al. 2020; Bouwman and Johnson 2018). We also explicitly consider the Big4 banks as a distinct group (see, e.g., Gete and Reher 2018). Our measures of the degree of regulatory shocks are changes to capital ratios, mortgage servicing rights, and lawsuit exposure.

Figure 2 reveals a stark contrast in the regulatory incidence on the Big4 relative to banks in all other size categories. During our sample, the Federal Reserve Board proposed and finalized rules that increased the risk weighting of holding mortgage servicing rights (MSR) on balance sheet. Panel 2a shows that prior to these changes (in 2008), mortgage servicing rights (as a fraction of tier 1 capital) were substantially larger for the Big4, even relative to other large banks. As a broader measure of the impact of MSR and other capital

⁸ See goodjobsfirst.org/violation-tracker.

⁹ We use all loans to capture a broader measure that can also allow for lenders to substitute across different loan programs (e.g., the decision to securitize or to hold on balance sheet).

regulation (including designation as a globally systemically important banks), we study regulatory capital in panel 2b: the Big4 started from the lowest level, and they had to increase their tier 1 capital ratios the most from 2008 to 2015. In panel 2c, we find that the fines arising from mortgage lawsuits are an order of magnitude larger for the Big4 compared to large banks. Even when scaled by total assets to account for the size of the Big4 (each of which is over \$1 trillion), panel 2d shows a clearly increasing relationship between size and the incidence of the penalties, with the Big4 hit disproportionately hard.

In sum, while the shocks to regulatory burden following the crisis were economically significant for all traditional banks, the evidence shows the incidence was higher for larger banks and particularly severe for the Big4. These new facts correlate with aggregate changes in market share shown in Figure 1: a massive decline by the Big4, followed by sizable declines for large and intermediate banks, while small banks enjoyed a slight growth in aggregate market share. Figure 2 strongly suggests that the relatively lighter touch of postcrisis regulation on small banks may play an important role in their relevance in a rapidly changing postcrisis landscape.

4. Big4 Retreat and Changing Lender Composition

Following the disproportionately large regulatory shock to the Big4, they had negative loan growth in virtually all counties. How do other classes of lenders respond? The aggregate patterns in Figure 1 suggest that there may be a pure substitution between Big4 and nonbanks. Table 2 presents the correlation matrix of county-level loan growth from 2008 to 2015 across lender classes (Big4, large banks, intermediate banks, small banks, credit unions, shadow banks, fintech) to allow for an initial examination of the local substitution patterns. Within banking, we broadly see negative correlations in growth for nearly all pairs of bank sizes, with the negative correlation between the Big4 and small

Table 2
Lender class growth rate correlations

	Big4	Large banks	Interm banks	Small banks	Credit union	Shadow banks	Fintech
Big4	1.00						
Large banks	−0.10	1.00					
Intermediate banks	−0.03	0.08	1.00				
Small banks	−0.27	−0.01	−0.11	1.00			
Credit unions	−0.03	0.03	−0.08	0.05	1.00		
Shadow banks	−0.17	0.09	−0.05	0.07	0.09	1.00	
Fintech	−0.12	0.13	−0.00	0.17	0.06	0.27	1.00

This table presents the correlation matrix of county-level loan growth for each lender class from 2008 to 2015. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo, *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018). GSE loans are loans sold to Fannie Mae or Freddie Mac.

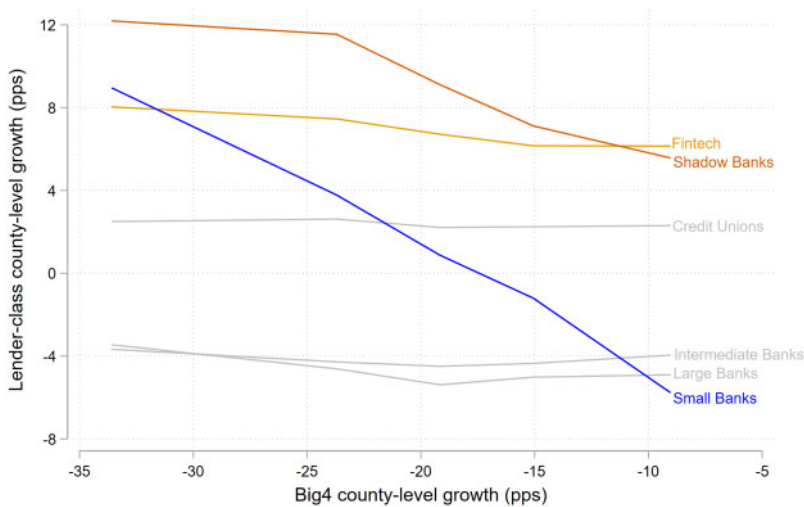


Figure 3
Big4 retreat and county-level lender class growth

This figure presents the county-level lender class growth from 2008 to 2015 as a function of the Big4 lending growth over the same period. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018).

banks being the largest at -0.27 . The only exception to these negative within-banking correlations is the positive comovement of large and intermediate banks (correlation = 0.08). The strong negative correlation between the Big4 and small banks even exceeds the correlation with shadow banks (-0.17) and fintech lenders (-0.12), who themselves are highly correlated with each other (0.27).

In Figure 3, we narrow our focus to the responses across lender classes to the Big4. We graphically present the county-level average loan growth rates for each lender class across quintiles of the Big4 growth rate. We first observe that Big4 growth is negative for all quintiles, with the top quintile having -9% growth. Figure 3 shows a large negative relationship between the loan growth by small banks and the Big4: there is a 14-pp difference in small bank lending growth between the bottom ($+8\%$) and top (-6%) quintiles of Big4 growth counties. The respective difference for shadow banks is about 6 pp, while all other lender classes appear relatively insensitive to the degree of Big4 withdrawal. Observing the average levels of growth in this figure reveals that both shadow banks and fintech lenders enjoyed strong positive growth across all quintiles. This widespread growth translates to the significant aggregate rise in nonbank lending (Figure 1). In contrast, the small bank growth across quintiles shows growth is not uniformly positive, which connects to the aggregate result in Figure 1 showing only modest small bank growth in the aggregate.

Appendix Table A.1A shows how the counties in the quintiles of high and low Big4 retreat differ on observable dimensions. The Big4 retreat was larger in areas with slightly higher minority populations, while none of the other observable characteristics shows a consistent pattern. With respect to the Big4's initial 2008 market share, Appendix Table A.1B shows that the Big4 typically had larger shares in areas with higher populations, higher minority shares of the population, and higher incomes. We control for these observable characteristics in all of our regressions.

4.1 Local response to the Big4 retreat: Regression estimates

We now estimate these relationships using linear regression. We separately estimate regressions of county-level growth in lending for each lender class during 2008–2015 on the county's contemporaneous growth in Big4 lending activity:

$$Growth_{county}^{LenderClass} = \phi \left(Growth_{county}^{Big4} \right) + \zeta_{stateFE} + \Gamma X_{county} + \epsilon_{county}. \quad (3)$$

Since these regressions are growth on growth, they minimize concerns about any time-invariant factors that might affect lender presence, baseline demand conditions, or local lending profitability. We include state fixed effects to control for any state-level differences in trends, such as legal environment or other state-specific unobserved drivers of growth rates across lender classes. We control for an extensive set of county-level factors X_{county} , including demographics (population, minority share, per capita income, age, subprime share), county house price growth, and local lending market conditions (banking deposits HHI, number of banks with a physical branch, and the number of traditional banks and number of nonbank lenders with mortgage presence). We also include the overall traditional bank share of mortgage lending in 2008 as a proxy for heterogeneity in demand for banking services across counties. Table 3 shows summary statistics of these characteristics for 2008. Our goal is to estimate the presence and extent of heterogeneity in response to the Big4, while controlling for local factors that could drive variation in demand. We cluster standard errors at the metropolitan statistical area (MSA) level to account for local spatial correlations beyond the county level.

Panel A of Table 4 presents the results, with each column representing a different lender class. Column 1 indicates small banks are highly sensitive to the Big4 growth: a 10-pp lower Big4 growth corresponds to a 6.6-pp increase in lending growth of small banks during the time period. The regression estimates also indicate that small banks' response to the Big4 retreat is stronger than that of shadow banks and fintech lenders, for whom a 10-pp lower Big4 growth corresponds to 1.3- and 0.5-pp higher growths, respectively (columns 2 and 3). Thus, while nonbank lenders experienced broad, secular growth, local growth was much less sensitive to county-level variation in the degree of Big4 withdrawal. Columns 4–6 show that the estimated relationships are either insignificant or not economically meaningful for large banks, intermediate

Table 3
County summary statistics

	Mean	SD	Min	p25	p50	p75	Max	N
<i>A. County characteristics</i>								
Population	89.24	185.45	0.98	12.38	27.14	69.44	1,211.10	2,983
Minority	0.21	0.19	0.01	0.05	0.13	0.32	0.97	2,983
Income	33.65	9.23	14.78	27.86	32.02	37.19	156.20	2,983
Subprime	0.33	0.09	0.08	0.26	0.32	0.40	0.65	2,983
Age	39.39	4.73	25.10	36.60	39.30	42.40	50.80	2,983
HPgrowth (%)	-2.14	12.28	-26.74	-10.49	-3.06	5.22	43.87	2,983
HHI	0.30	0.19	0.05	0.17	0.25	0.38	1.00	2,983
Banks (physical)	9.11	9.96	1.00	4.00	6.00	11.00	163.00	2,983
Banks (lending)	46.01	32.50	4.00	26.00	37.00	56.00	363.00	2,983
Nonbanks (lending)	51.33	44.39	1.00	20.00	38.00	67.00	308.00	2,983
<i>B. 2008 county mortgage lending shares</i>								
Big4	27.13	11.19	0.00	18.88	25.77	33.65	80.00	2,983
Large Banks	12.13	8.37	0.00	6.04	10.38	16.27	58.40	2,983
Intermediate Banks	8.94	6.44	0.00	4.52	7.89	11.61	52.60	2,983
Small Banks	30.10	17.72	0.00	15.60	27.69	42.25	87.50	2,983
Credit Unions	5.20	6.05	0.00	1.20	3.22	7.08	57.66	2,983
Shadow Banks	14.88	8.20	0.00	8.83	13.89	19.52	57.25	2,983
Fintech	1.63	1.63	0.00	0.69	1.25	2.08	21.05	2,983

This table presents county-level summary statistics for 2008. *Population* is the population in thousands; *Minority* is the non-white share of the population; *Income* is the per capita income in thousands; *Subprime* is the share of the population with FICO below 660; *HPgrowth* is the county-level house price growth during 2008–2015; *Age* is the median age of residents; *HPgrowth* is the growth in house prices during the sample; and *HHI* is the county banking competition based on deposits. *Banks (physical)* is the number of unique banks with a branch in the county; *Banks (lending)* is the number of unique banks with mortgage lending activity in the county; and *Nonbanks (lending)* is the number of unique nonbanks (shadow banks and fintech) with mortgage lending activity in the county. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018).

banks, or credit unions. As shown earlier in Section 3, large and intermediate banks faced regulatory burden increase in a range between small banks and the Big4, so the lack of strong substitution is unsurprising. The lack of a strong relationship to credit unions is likely attributable to the institutional features that can constrain credit union lending expansion (e.g., borrowers belonging to a particular profession).

In sum, these results on local substitution indicate that aggregate patterns in lending activity mask significant heterogeneity at the local level. Rather than mirroring the aggregate pattern of nonbanks substituting for the Big4, the county-level analysis shows a sizable within-traditional-bank substitution between the Big4 and small banks. Thus, amid the secular increase in nonbank lending, this reshuffling of lending within traditional banks plays an important role in explaining small banks' persistence in this market.

4.2 Local response to the Big4 retreat: IV estimates

To conclusively interpret the estimated coefficients above as supply side substitutions across lender classes, the ideal experiment would randomly assign

Table 4
Cross-sectional heterogeneity in response to the Big4 retreat

	(1) Small banks	(2) Shadow banks	(3) Fintech	(4) Large banks	(5) Interm banks	(6) Credit unions
$Growth_{county}^{Big4}$	−0.659*** (<0.01)	−0.130*** (<0.01)	−0.047*** (<0.01)	−0.040*** (0.01)	0.002 (0.91)	−0.010 (0.41)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,983	2,983	2,983
R^2	.295	.418	.232	.309	.258	.224

p-values are in parentheses.
* $p < .10$; ** $p < .05$; *** $p < .01$.
This table presents OLS estimates from the regressions of county-level growth in mortgage originations for each lender class from 2008 to 2015 on the county-level growth in Big4 originations during this time period ($Growth_{county}^{Big4}$). *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018); *Large banks* includes banks with assets between \$50 billion and \$1 trillion; and *Intermediate Banks* includes banks with assets between \$10 billion and \$50 billion. Control variables include 2008 county-level logarithm of the 2008 population, non-white share of the population, per capita income, median age, share of the population with FICO below 660, county banking HHI based on deposits, house price growth, the logarithm of the number of banks with a physical branch in the county, the logarithm of the number of banks with at least one mortgage loan in the county, and the logarithm of the number of shadow banks or fintech lenders with at least one mortgage loan in the county. All regressions include state fixed effects. Standard errors are clustered by MSA.

Big4 growth rates across counties and observe the responses of the other lenders. Since such randomization is impossible, the presence of unobserved demand shocks is a potential challenge to our interpretation. Specifically, our results would be confounded if county-level drops in demand for mortgage products or bundled services supplied by the Big4 in 2008–2015 are correlated with an increase in demand for small bank products or services during the same period. Though our control variables include changes in house prices, demographics, and a number of lending market characteristics that should help account for heterogeneity in county-level conditions, we now use an instrument variables approach for the Big4 withdrawal to further abstract away from such confounding factors.

The first of our two main instruments for the county-level Big4 withdrawal is the initial condition of the Big4’s county-level lending share in 2008. The instrument does not condition on the actual withdrawal—which could be correlated with unobserved demand shocks during 2008–2015—instead it identifies counties in which the Big4 had a larger presence and thus a larger scope for withdrawal.¹⁰ We view this IV as a reduced-form measure that captures multiple regulatory shocks (e.g., G-SIB designation, stress tests, lawsuits, capital constraints) that particularly affected the Big4. Before the

¹⁰ Recent work has used the local banking market conditions prior to the Dodd-Frank Act as an instrument for future levels of lending (see, e.g., Gete and Reher 2018; D’Acunto and Rossi 2022). Although our approach is somewhat similar, we instrument for future *growth* rather than future levels.

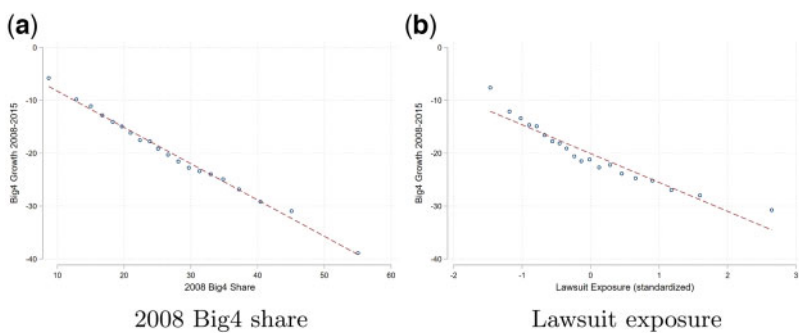


Figure 4
Instruments for Big4 growth during 2008–2015
This figure presents a bin scatter plot of county-level growth for the Big4 from 2008 to 2015 (y-axis) against the 2008 Big4 county share of mortgage originations (panel a) and standardized lawsuit exposure (panel b). *Big4* includes Bank of America, Citi, JP Morgan, and Wells Fargo.

postcrisis retreat of Big4 from mortgage lending, there was a great deal of variation in their origination share across the country. In 2008, the Big4 accounted for about 27% of new loans in the average county, with a range of 14% to 42% share from the 10th to 90th percentile. The average county had Big4 growth of about -20% with the 10th to 90th percentile range of -31% to -10% , with no counties experiencing positive growth. Our instrument exploits the fact that the counties in which the Big4 had the largest initial presence are the counties with the greatest scope for meaningful withdrawal. Figure 4a graphically shows this strong “first-stage” relationship. A 10-pp higher 2008 Big4 share in a county is associated with an economically and statistically significant 6.9-pp lower growth in Big4 lending from 2008 to 2015.

Figure 5 shows the geographical variation in Big4 share in terms of their initial 2008 dispersion (panel 5a) and variation in the changes in Big4 share during their retreat from this market during 2008–2015 (panel 5b). Panel 5a highlights the relative dominance of the origination markets in the West, in Florida, and in parts of the East Coast as compared to the Midwest and South. Panel 5b shows the county-level variation in the gap the Big4 left when they retreated from the market during 2008 to 2015. While Figure 5 shows substantial regional variation, note that the tests difference out these broad patterns and examine the changes in lender-class behavior within a state.

We estimate the following two-stage least squares regressions for each respective lender class, with our objective being to examine which lender classes are locally stepping in to serve the marginal borrowers that otherwise would have been served by the Big4.

$$Growth_{county}^{Big4} = \theta Share_{county}^{08Big4} + \zeta_{stateFE} + \Gamma X_{county} + \eta_{county} \tag{4}$$

$$Growth_{county}^{LenderClass} = \psi \left(\widehat{Growth_{county}^{Big4}} \right) + \xi_{stateFE} + \Lambda X_{county} + \epsilon_{county} \tag{5}$$

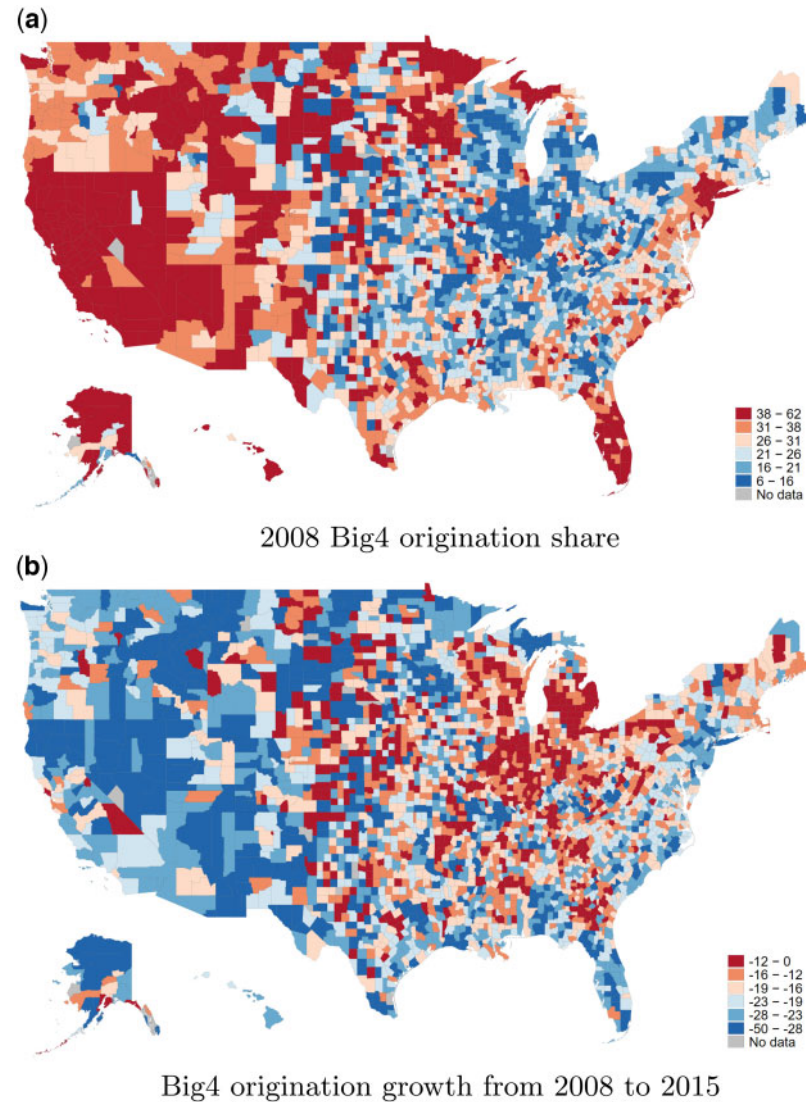


Figure 5
Mapping 2008 Big4 share and 2008–2015 Big4 growth
This figure presents the share of mortgages by the Big4 banks (Bank of America, Citi, JP Morgan, and Wells Fargo) in 2008 in panel a. Panel b plots the growth in the Big4 originations from 2008 to 2015.

Panel A of Table 5 presents the results. The first-stage F-statistic is very strong (over 2,000), which reflects the large and widespread nature of their retreat (see also its graphical representation of the first stage in Figure 4a). Column 1 indicates that a 10-pp larger decrease in $Growth_{county}^{Big4}$ corresponds

Table 5
Cross-sectional heterogeneity in response to the Big4 retreat: IV regressions

	(1) Small banks	(2) Shadow banks	(3) Fintech	(4) Large banks	(5) Interm banks	(6) Credit unions
<i>A. 2008 Big4 share IV estimates</i>						
$Growth_{county}^{Big4}$	−0.884*** (<0.01)	−0.259*** (<0.01)	−0.122*** (<0.01)	−0.079*** (<0.01)	−0.003 (0.84)	−0.013 (0.49)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,983	2,983	2,983
R ²	.148	.147	.050	.034	.027	.024
<i>B. Lawsuit IV estimates</i>						
$Growth_{county}^{Big4}$	−0.970*** (<0.01)	−0.353*** (<0.01)	−0.173*** (<0.01)	−0.066** (0.02)	−0.012 (0.62)	−0.019 (0.34)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,983	2,983	2,983
R ²	.139	.132	.016	.035	.027	.024

p-values are in parentheses.
* $p < .10$, ** $p < .05$; *** $p < .01$.

This table presents IV estimates from the regressions of county-level growth in mortgage originations for each lender class from 2008 to 2015 on the county-level growth in Big4 originations during this time period ($Growth_{county}^{Big4}$). *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018); *Large banks* includes banks with assets between \$50 billion and \$1 trillion; and *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion. $Growth_{county}^{Big4}$ is instrumented with the 2008 county-level origination share of the Big4 (panel A) and with the standardized exposure to postcrisis mortgage lawsuit penalties (panel B) as defined in Equation (6) in the main text of the paper. Control variables include 2008 county-level logarithm of the 2008 population, non-white share of the population, per capita income, median age, share of the population with FICO below 660, county banking HHI based on deposits, house price growth, the logarithm of the number of banks with a physical branch in the county, the logarithm of the number of banks with at least one mortgage loan in the county, and the logarithm of the number of shadow banks or fintech lenders with at least one mortgage loan in the county. All regressions include state fixed effects. The F-statistics are 2,670 and 957 for the first-stage regression for panels A and B, respectively. Standard errors are clustered by MSA.

to a 8.8-pp increase in the lending growth of small banks, while the estimates are much smaller for nonbanks at 2.6 and 1.2 pp for shadow banks and fintech lenders, respectively. Similar to the OLS estimates, we find the IV estimates for the large banks, intermediate banks, and credit unions to be economically small.

The exclusion restriction in this IV setup requires that the 2008 Big4 county-level market presence be related to small bank growth from 2008 to 2015 only through its effect on Big4 2008–2015 growth. A violation of the exclusion restriction would occur if the 2008 Big4 shares are correlated with county-level unobservables (e.g., changes in demand for different types of bank services) that independently drive future small bank growth. We note that all of our regressions explicitly control for several variables that could be correlated with unobserved demand shocks, including house price changes, demographics, measures of average creditworthiness, and local market conditions, including the total traditional banks’ share of mortgage lending. Second, we reestimate our regressions using further lags of our instrument. We also repeat our IV

estimation using Big4 market presence in 2005 and then again using their presence in 2002 as instruments for the Big4 postcrisis retreat. These lagged instruments abstract away from demand shocks potentially related to the Big4's 2008 local market presence and future small bank growth. It is less likely that unobservable local conditions correlated with past Big4 market presence can consistently explain changes in small bank growth well into the future (2008–2015). By using Big4 2002 levels as an instrument, we even precede the major portion of the boom in house prices, yet we find consistent results. Appendix Table A.2 presents the estimates. Thus, in the absence of the ideal experiment, these tests lend support to the credibility of our interpretation.

Next, we use an instrument that explicitly links a key aspect of postcrisis regulatory burden to changes in Big4 lending. In particular, we construct an instrument that exploits the disproportionate incidence of postcrisis lawsuits on the Big4 as discussed in Section 3. Similar to Buchak et al. (2018), we quantify the county-level (c) exposure to lawsuit shocks, where L_b is the accumulated lawsuits during 2008–2015 for bank b .

$$Lawsuit_c = 100 \times \sum_{b \in c} L_b \frac{Originations_{bc2008}}{\sum_{d \in c} Originations_{dc2008}} \quad (6)$$

Panel B of Table 5 presents the results. The first-stage F-statistic is very strong (over 900), and we show this relationship graphically in Figure 4b. Column 1 indicates that a 10-pp larger decrease in $Growth_{county}^{Big4}$ corresponds to a 9.7-pp increase in the lending growth of small banks, while the estimates are much smaller for nonbanks (columns 2 and 3). Overall, the estimates using the lawsuit exposure are similar and slightly larger than our earlier results using the 2008 Big4 share as the instrument. The exclusion restriction for the lawsuit IV requires that lawsuits accumulated over 2008–2015 affect small bank lending growth from 2008 to 2015 only through their effect on Big4 growth rates. The nature of lawsuits lends credence to the exclusion restriction: the largest lawsuits often particularly targeted the Big4 and represented nationwide, bank-level shocks. In contrast, small banks were virtually untouched (Figure 2c). In sum, we find similar results for the strong within-traditional banking substitution between the Big4 and small banks using both the broader (2008 Big4 share) and the more direct (lawsuit exposure) instruments for the Big4 retreat.

To further support the regulation channel as a key explanation for small bank persistence, we perform an additional test that exploits variation in regulatory shocks within the largest four lenders. This test estimates small bank lending sensitivity to the Big4 member hit hardest by the regulatory shocks and allows us to hold fixed the drivers of ex ante Big4 share. We find that even after accounting for the baseline heterogeneity in ex ante Big4 shares, small bank lending grew significantly more in areas in which Big4 constituents were hit harder with regulatory shocks. Appendix A discusses the details of this test.

4.3 Reallocation of lending within small banks

So far, we have shown that local small bank response to the Big4 retreat is strongest among all lender classes. At the same time, Figure 1 shows small banks experienced only a modest *aggregate* increase in market share. Appendix B delves deeper into how small banks reallocate their lending at the individual bank level. We briefly summarize the results of that investigation here. We find the small bank substitution is driven by a combination of within-lender reallocation and differences in growth across banks. A given small bank is more likely to grow their lending in the parts of their geographical footprint where the Big4 growth was lower. On the extensive margin, the new counties that a given small bank enters are those with relatively lower Big4 growth. When the data are collapsed to the bank level, those whose lending footprint experienced a greater Big4 withdrawal have higher bank-level lending growth. The relative size of the estimates for within-bank and across bank variation (Appendix Table A.4) suggests that differences in exposure across small banks plays a larger role than within-small-bank reallocation. Thus, whether an individual small bank experienced positive or negative lending growth was a function of the exposure of their respective geographical footprints to Big4 growth. This stands in contrast with nonbanks, whose growth was more uniform across the country. In the next subsection, we tackle how the ability to lend on balance sheet plays a role in explaining these patterns.

4.4 Balance sheet lending

A key distinction between banks and nonbanks is the ability to lend on balance sheet. In this subsection, we examine whether this similarity among banks paired with the disproportionate shocks to form a key channel for the small banks' substitution with the Big4. As small banks also experience an absolute increase in regulation, it is not obvious whether this channel will be at play. To directly test for a greater small bank lending response in balance sheet lending compared to non-balance-sheet lending, we use a stacked regression where the level of observation is now at the county $\times$ loan segment level. We define the loan segments as balance sheet lending and non-balance-sheet lending.¹¹ That is, we have two observations for each county in the sample: one with balance sheet lending growth ($1[\text{BS-lending}] = 1$ in the regressions below) and one with non-balance-sheet lending growth. We can then estimate a modified version of our main IV regression for each lender class that allows us to estimate the incremental sensitivity of balance sheet lending growth relative to sensitivity to non-balance-sheet lending growth. Specifically, we estimate the following specification, where the first two equations are the relevant first stages followed by the second stage regression. The control variables are the same as the baseline

¹¹ As discussed in Section 2, we categorize loans as "on-balance-sheet" if they are coded as "not sold" or "sold to an affiliate" in the HMDA data set.

Table 6
Substitution in balance sheet lending

	(1) Small banks	(2) Large banks	(3) Intermediate banks	(4) Shadow banks	(5) Fintech	(6) Credit Unions
$\widehat{Growth_{county}^{Big4}}$	−0.567*** (< 0.01)	−0.104*** (< 0.01)	−0.012 (0.64)	−0.375*** (< 0.01)	−0.123*** (< 0.01)	0.016 (0.56)
$\widehat{Growth_{county}^{Big4} \times 1[BS-lending]}$	−0.580*** (< 0.01)	0.063 (0.10)	−0.002 (0.95)	0.422*** (< 0.01)	0.100*** (< 0.01)	−0.065 (0.13)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,954	5,954	5,954	5,954	5,954	5,954
R ²	.088	.019	.015	.061	.023	.014

p-values are in parentheses.
* *p* < .10; ** *p* < .05; *** *p* < .01.
This table presents IV estimates from the regressions of county-level growth in balance sheet or non-balance-sheet mortgage originations for respective lender classes from 2008 to 2015 on the county-level growth in Big4 originations during this time period. Each county has two observations: one for the lender class's growth in balance sheet lending (indicated by 1[BS-lending]) and one for their growth in non-balance-sheet lending (the omitted category). $\widehat{Growth_{county}^{Big4}}$ represents the 2008–2015 loan growth for the Big4. The instruments for $\widehat{Growth_{county}^{Big4}}$ and $\widehat{Growth_{county}^{Big4} \times 1[BS-lending]}$ are the 2008 county-level share of the Big4 and its interaction with 1[BS-lending]. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018). Control variables are the same as those used in Table 4, and the regressions also include state×loan segment (balance sheet vs. non-balance-sheet) fixed effects. The Cragg-Donald F-statistic is 602. Standard errors are clustered by MSA.

specification (3) and state×1[BS-lending] fixed effects:

$$Growth_{county}^{Big4} = \gamma Share_{county}^{08Big4} + Controls + v_{county}, \tag{7}$$

$$Growth_{county}^{Big4} \times 1[BS-lending] = \theta Share_{county}^{08Big4} + Controls + \eta_{county}, \tag{8}$$

$$\begin{aligned} Growth_{county}^{LenderClass} = & \psi \widehat{Growth_{county}^{Big4}} + \\ & \phi \widehat{Growth_{county}^{Big4} \times 1[BS-lending]} \\ & + Controls + \epsilon_{county}. \end{aligned} \tag{9}$$

The point estimate on ψ represents the degree of substitution between the Big4 and small banks in non-balance-sheet lending, and the point estimate on the interaction (ϕ) estimates the incremental sensitivity in balance sheet lending beyond the sensitivity in non-balance-sheet lending.

Table 6 presents the results for each lender class, with column 1 presenting the estimates for small banks. The point estimate on Big4 growth ($\hat{\psi} = -0.567$) shows a strong response for small bank non-balance-sheet lending. The estimate for the interaction term ($\hat{\phi} = -0.580$) is economically and statistically significant and indicates that small bank balance sheet lending is roughly twice as responsive as non-balance-sheet lending. The sum of the two coefficients

of ($\hat{\psi} + \hat{\phi} = -1.147$) reflects the overall substitution for balance sheet lending. We do not find differential sensitivity for large and intermediate banks, which likely reflects the relatively greater balance sheet pressure during the sample relative to small banks. For shadow banks and fintech lenders in columns 4 and 5, we observe that their sensitivity is concentrated in non-balance-sheet lending and that their the overall sensitivity of balance sheet lending (sum of the coefficients) is close to zero, which is consistent with their business model.

These results shed light on a key source of competitive advantage for small banks. This helps explain why small bank lending is relatively more responsive compared to their even more lightly regulated nonbank competitors, which lack the ability to seriously compete in balance sheet lending. Note that the strong substitution in balance sheet lending between the largest and smaller banks occurs amid absolute increases in regulation for traditional banks of all sizes. Regulatory pressure on bank balance sheets is reflected in the negative aggregate growth of small bank balance sheet lending. At the same time, small banks experienced aggregate positive growth in GSE lending that was stronger than all other traditional banks. We examine this fact more closely in the next section, where we study a second major shock to the supply side that sheds new light on small banks' lending dynamics.

5. Shocks to the Cost of GSE Lending

To shed further light the continuing importance of small banks in the postcrisis mortgage market, we now examine significant changes in GSE financing costs that created distinct opportunities for small banks. During the second half of our sample period, there were large changes in the cost schedule for loans that lenders wish to sell to the GSEs (Fannie Mae and Freddie Mac). The GSEs purchase loans from lenders and create mortgage-backed securities primarily to sell to investors.¹² The lenders pay guarantee fees (g-fees) to the GSEs to account for the default risk of the loans they sell them. These fees allow the GSEs to guarantee principal and interest payments (i.e., eliminate default risk) to the investors. From 2011, the Federal Housing Finance Agency (FHFA) required that the GSEs increase their g-fees to limit taxpayer losses and reduce the GSEs' dominant position in the housing market. The size of the increases in g-fees were substantial and varied across lenders according to volume of loans sold to the GSEs.

Figure 6 shows the changes in g-fees for the largest GSE lenders and small GSE lenders for 2008–2015. The figure highlights two prominent changes: (1) a transition to parity in fees paid across lenders of different sizes and (2) a sizable increase in the level of g-fees paid across all lenders. We discuss the implications of these changes below, focusing on how these may shed light on

¹² The GSEs can also bundle a lenders loan into an MBS that the lender will receive, often called the "swap" program. Other loans may remain on the GSEs' balance sheets.

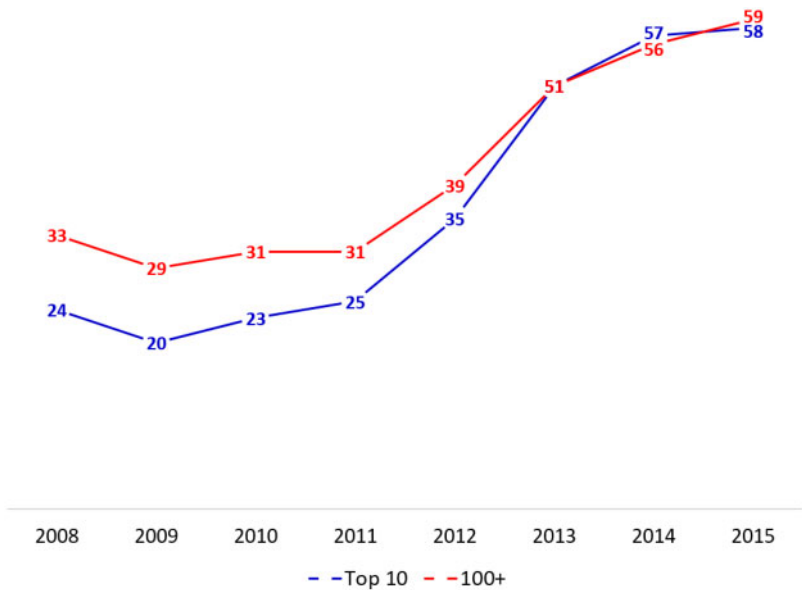


Figure 6
Change in the parity and levels of guarantee fees
This figure presents the average level of guarantee fees (in basis points) paid by lenders selling loans to the GSEs (Fannie Mae and Freddie Mac) over 2008–2015. We separately plot the g-fees paid by the top-10 GSE lenders and the g-fees paid by lenders outside the top-100 lenders.

the dynamics of small banks’ lending in the mortgage market. In this portion of the analysis, we narrow our focus to a tighter window of 2011–2015, when these changes were taking place.

5.1 G-fee shocks and small bank growth in GSE lending

Small banks’ average GSE lending growth of 5% during this period was substantially larger than other traditional banks: the average growth for the Big4, large banks, and intermediate banks was -20%, -8%, and 1%, respectively (Table 1). One reason this growth is striking is that these loans are securitized, so the differential balance sheet pressures across bank size that affects loan retention are unlikely to explain this pattern.

Starting in 2011, the GSEs erased the cost advantage that larger, high-volume lenders had previously enjoyed relative to smaller lenders.¹³ This relative

¹³ A report by the Office of the Inspector General summarizes the reason of the change as follows: “As part of its August 2012 directive to raise guarantee fees by 10 basis points, FHFA also required the Enterprises to aim for uniform pricing across all lenders, regardless of their volume of transactions, by applying differential increases for swaps and cash purchases designed to reduce significantly the previous pricing difference between large and small lenders, on average. FHFA said it took this step to help eliminate cross-subsidies between lenders as well as between mortgage products. For example, to add uniformity to the fees the Enterprises charge lenders, the increase would be larger for lenders delivering larger volumes of mortgages than for lenders delivering smaller

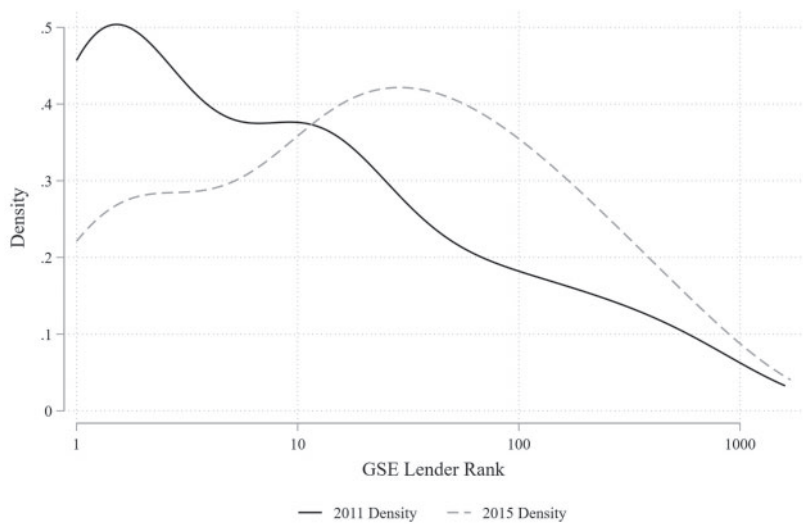


Figure 7
Changing distribution of GSE lending

This figure presents the density of GSE lending according to the rank of the lender. The GSE Lender Rank is computed annually by the total number of loans sold to the GSEs (which includes originations and loans purchased and subsequently sold to the GSEs within the year). We present the densities for 2011 and 2015.

increase in the cost to larger lenders for GSE lending leveled the playing field for smaller lenders. To illustrate the subsequent changes in GSE lending across lender size following the cost shock, Figure 7 plots the densities of GSE lending in 2011 and 2015 by the rank of the lender (by GSE loan volume) that year. The figure shows a significant transition from a market dominated by the top lenders in 2011 to a flatter distribution with much more GSE lending done by small to midsize lenders. Figure 7 suggests the move toward parity in g-fees may be a key factor for the growth of small banks in the GSE segment.

To formally test this possibility, we test whether small bank growth in GSE lending varied according to how much a county was exposed to the parity shock. Specifically, we regress small bank county-level GSE lending growth from 2011 to 2015 on the degree to which the county was dominated by high-volume GSE lenders whose high-volume g-fee discount prior to the parity shock was erased. We capture the extent of this impact by the origination share of top-10 GSE lenders in 2011 ($Top10GSELenderShare_{county}^{2011}$). While the choice of using the

volumes” (FHFA “Initiative to Reduce the Enterprises’ Dominant Position in the Housing Finance System by Raising Gradually Their Guarantee Fees”).

Table 7
Shocks to the costs of GSE lending and small bank growth

Lender class: Loan Segment:	Parity		Level		Validation	
	(1) Small bank GSE	(2) Small bank GSE	(3) Small bank balance sheet	(4) Small bank balance sheet	(5) Small nonbank GSE	(6) Small bank jumbo
Top10GSELenderShare2011	0.230*** (< 0.01)	0.429*** (< 0.01)		0.524*** (< 0.01)	0.128** (0.04)	0.098 (0.65)
GSE share 2011		-0.729*** (< 0.01)	0.515*** (< 0.01)	0.344*** (< 0.01)	-0.458*** (< 0.01)	0.120 (0.71)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,966	2,966	2,973	2,973	2,966	1,748
R ²	.020	.103	.035	.061	.127	.012

p-values are in parentheses.
* p < .10; ** p < .05; *** p < .01.
This table presents estimates from the regressions of county-level small bank lending growth in mortgage originations from 2011 to 2015 on measures of exposure to the g-fee parity and level shocks. The dependent variables are small bank growth in GSE lending (columns 1 and 2), small bank growth in balance sheet lending (columns 3 and 4), and small bank growth in jumbo lending (column 6). The dependent variable in column 5 is small nonbank growth in GSE lending, where small nonbanks are defined as nonbanks outside of the top-10 volume lenders. *Top10GSELenderShare2011* represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year. *GSE Share 2011* represents the share of all county loans in 2011 that were sold to the GSEs that year. Control variables are the same as those in Table 4 and include state fixed effects. Standard errors are clustered by MSA.

top 10 is somewhat arbitrary, the results are not sensitive to this selection.¹⁴

$$GSE\ Growth_{county}^{SmallBank} = \pi \left(Top10GSELenderShare_{county}^{2011} \right) + \zeta_{stateFE} + \Gamma X_{county} + \epsilon_{county} \tag{10}$$

Column 1 in Table 7 presents the estimates. Small banks increased their GSE lending more in counties in which their incumbent competitors were more affected by the parity shock. A 10-pp higher *Top10GSELenderShare*²⁰¹¹ corresponds to 2.3-pp higher growth in small bank GSE lending. In column 2, we control for the initial share of the loans in the county that were GSE loans (*GSE Share 2011*). After accounting for this baseline heterogeneity in the local composition of loan types, we find that a 10-pp higher *Top10GSELenderShare*²⁰¹¹ leads to a 4.3-pp higher growth in small bank GSE lending. In sum, the transition to g-fee parity leveled the playing field in the GSE market for small lenders. Small banks, in turn, increased their GSE lending broadly across the country, and particularly in areas in which leading incumbent lenders were more affected by the g-fee parity shock.

5.2 G-fee shocks and small bank growth in balance sheet lending

In this subsection, we exploit that, in addition to g-fee parity between small and large lenders, there were large increases to the *level* of g-fees from 2011 to

¹⁴ Also, in untabulated tests, we estimate the following using the top lenders' 2008 county-level share of lending as an instrument for *Top10GSELenderShare*²⁰¹¹_{county}. We find similar results.

2015. Figure 6 shows that the average levels of g-fees doubled from about 30 to 60 bps. In light of this substantial increase in the cost of doing business with the GSEs, banks—in contrast with nonbanks—have another potential margin of adjustment: retention of loans on their balance sheet (Buchak et al. 2020). Thus, the hike in g-fees can open up growth opportunities for on-balance-sheet lending.

We hypothesize that these opportunities for balance sheet lending growth of small banks grows will be greater in areas previously characterized by more GSE lending (i.e., higher *GSE Share 2011*) and areas previously dominated by the top GSE lenders (who experience a disproportionate shock to the g-fee levels). We use a similar framework as in Equation (10), except now we examine whether the balance sheet lending of small banks grows more in areas most affected by the shock to the level of the g-fees.

Column 3 of Table 7 shows that areas with a 10-pp higher *GSE Share 2011* experienced 5.2-pp higher small bank balance sheet lending growth. Column 4 also includes *Top10GSELenderShare*²⁰¹¹, which represents areas dominated by lenders who experience the greatest increase in g-fee costs. The results show that balance sheet lending by small banks grows most in areas previously dominated by GSE lending overall and especially areas previously dominated by the top GSE lenders. A 10-pp higher *Top10GSELenderShare*²⁰¹¹ corresponds to a 5.2-pp higher small bank balance sheet lending growth rate, and 10-pp higher *GSE Share 2011* corresponds to a 3.5-pp higher small bank balance sheet lending growth rate. These results on growth in balance sheet lending due to an increase in GSE financing costs are consistent with the theoretical prediction in Buchak et al. (2020).

5.3 Further validation of the G-fee shocks

We now present two additional tests to further validate the g-fee shock mechanism. First, we note that the transition to g-fee parity leveled the playing field in the GSE market for smaller *lenders* within the GSE segment, not just small banks alone. Thus, we would expect this shock to also present opportunities for GSE lending growth among small nonbank lenders. In column 5 of Table 7, we find evidence that smaller nonbanks indeed had higher growth in areas with more exposure to the “parity” shock (i.e., higher *Top10GSELenderShare*²⁰¹¹).

Finally, we present a placebo test to lend support to the argument that the g-fee changes are indeed driving our results. The changes in g-fees should affect lending activity for loans below the conforming loan limits because selling loans above that limit (jumbo loans) to the GSE is not possible. Column 6 of Table 7 examines the small bank lending growth in jumbo lending. We find that the variables proxying for shocks to g-fees are unrelated to the changes in jumbo lending, consistent with g-fee changes not affecting this market segment.

Table 8
Distinct role of the shocks to GSE funding

	All		GSE		Balance sheet	
	(1)	(2)	(3)	(4)	(5)	(6)
A. Top-10 GSE lender shares and small bank lending growth						
Top10GSELenderShare2011	0.519*** (< 0.01)		0.429*** (< 0.01)		0.524*** (< 0.01)	
Top10Share Big4		0.565*** (< 0.01)		0.504*** (< 0.01)		0.575*** (< 0.01)
Top10Share nonbank		1.294*** (< 0.01)		0.733** (0.02)		1.256*** (0.01)
Top10Share other bank		0.450*** (< 0.01)		0.324*** (0.01)		0.441*** (< 0.01)
GSE share 2011	0.026 (0.70)	0.011 (0.87)	-0.729*** (< 0.01)	-0.744*** (< 0.01)	0.344*** (< 0.01)	0.328*** (< 0.01)
	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,966	2,966	2,973	2,973
R ²	.103	.113	.103	.106	.061	.064
B. Top-50 GSE lender shares and small bank lending growth						
Top50GSELenderShare2011	0.497*** (< 0.01)		0.425*** (< 0.01)		0.548*** (< 0.01)	
Top50Share Big4		0.575*** (< 0.01)		0.514*** (< 0.01)		0.594*** (< 0.01)
Top50Share nonbank		0.680*** (< 0.01)		0.741*** (< 0.01)		0.770*** (< 0.01)
Top50Share other bank		0.420*** (< 0.01)		0.310*** (< 0.01)		0.465*** (< 0.01)
GSE Share 2011	-0.047 (0.47)	-0.056 (0.39)	-0.797*** (< 0.01)	-0.809*** (< 0.01)	0.249** (0.02)	0.249** (0.03)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,966	2,966	2,973	2,973
R ²	.135	.140	.120	.127	.077	.077

p-values are in parentheses.
* $p < .10$; ** $p < .05$; *** $p < .01$.
This table presents estimates from the regressions of small banks' county-level lending growth in mortgage originations from 2011 to 2015 on measures of exposure to the g-fee cost shocks. The dependent variables are small bank lending growth for all loans (columns 1 and 2), GSE loans (columns 3 and 4), and balance sheet loans (columns 5 and 6). Panel A (B) uses the top-10 (top-50) GSE lenders to measure the degree of g-fee shock. *Top10GSELenderShare*2011 represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year. *Top10Share X* represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year that are members of group *X*, where *X* is Big4, nonbank lenders, or non-Big4 banks. *GSE Share 2011* represents the share of all county loans in 2011 that were sold to the GSEs that year. The variables in panel B are defined analogously using the top-50 GSE lenders. Control variables are the same as those used in Table 4 and include state fixed effects. Standard errors are clustered by MSA.

5.4 GSE cost shocks versus differential incidence of regulation

Our results have emphasized two shocks to the supply side of mortgage lending that shed light on small bank lending dynamics: differential postcrisis banking regulation and changes to GSE financing costs. However, the fact that the Big4 are included in the top-10 GSE lender share (*Top10GSEShare*²⁰¹¹_{county}) may call into question whether our results in the previous section capture a distinct shock to the supply side of mortgage lending. While fully assessing the relative contributions of both changes would require a structural model that is beyond the paper's scope, we perform additional tests that further establish the distinct

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role played by the shocks to g-fees in explaining the persistence of small banks in the mortgage market.

In Table 8, we reestimate the relationship between $Top10GSEShare_{county}^{2011}$ and small bank lending growth for all loans, GSE loans, and balance sheet loans for reference in columns 1, 3, and 5. We then decompose the top-10 GSE lender group into three component parts: those that are Big4, nonbank top-10 GSE lenders, and other bank top-10 GSE lenders. In columns 2, 4, and 6, we present estimates of small bank lending growth for the different loan segments as a function of these three components of the top-10 GSE lenders. If the g-fee channel is distinct from the differential regulatory shock channel (both of which affect the Big4), then we would expect that the share of non-Big4 top GSE lenders also would be related to small bank lending growth. We find that the effect of g-fee increases on the top nonbank lenders has a particularly large effect on small bank balance sheet lending. Overall, the results support the idea that the g-fee channel is distinct from the differential impact of postcrisis regulatory burden.

To ensure our results are not driven by the particular choice of using the top-10 lenders to measure the degree of the shocks, we repeat these tests using the top-50 lenders in panel B of Table 8. We find similar results, which lends further support to the distinct contribution that the shocks to GSE funding costs play in explaining small bank lending.

6. Conclusion

The postcrisis era of mortgage lending has been characterized by large increases in bank regulation and a surge in shadow bank and fintech lending. However, we show that small bank lending has remained remarkably stable, even showing modest growth. Small banks' stability stands in stark contrast with the rest of traditional banks, whose market shares have declined in the years following the financial crisis. Our analysis sheds light on key supply side factors—the relatively lighter regulatory touch on small banks and changes to the cost structures of the mortgage market—that have allowed small banks to endure and remain important players in this market.

We first provide new evidence that key elements of regulatory burden, including capital regulation and postcrisis fines, increase with bank size and disproportionately affect Big4 banks. In light of this heterogeneity in the incidence of postcrisis regulatory burden, we document a strong within-traditional-banking substitution pattern between the withdrawing Big4 banks and small banks. In contrast to nonbanks, small banks' ability to lend on balance sheet is a key channel of this substitution.

As a second channel explaining small banks' lending dynamics, we provide novel evidence that changes to GSE financing costs were particularly advantageous to small banks. During 2011–2015, the guarantee fees (g-fees) paid by lenders to the GSEs were substantially increased, and the discount

given to high-volume lenders was eliminated. Small banks grew their GSE lending the most in areas previously dominated by lenders enjoying these volume-based discounts. The overall increase in the level of g-fees increased the costs of doing business for GSE lenders, effectively altering the relative attractiveness of securitization versus loan retention. Indeed, we find small banks grew their balance sheet lending more in areas most affected by the g-fee hikes.

Appendix A. Exploiting Within-Big4 Variation

To support the interpretation that disproportionate regulatory shocks to the Big4 led to strong substitution of small bank lending for the retreating Big4, we introduce a test that exploits variation in the regulatory shocks within the largest four lenders. Bank of America (BoA) was the hardest hit out of the Big4, alone accounting for over half of the total fines. They cut their origination activity from 2008 to 2015 by a staggering two-thirds. We exploit this heterogeneity in the size of the lawsuit shock within the Big4 in a stringent test of our regulatory burden hypothesis. We examine the sensitivity of small bank lending growth to BoA's retreat, while directly controlling for the initial Big4 share. This test allows us to exploit variation in the regulatory shock, while controlling for unobserved factors that could be correlated with the ex ante Big4 share. Said differently, our aim is to hold fixed the drivers of ex ante Big4 share, while estimating the incremental small bank lending sensitivity to the member of the Big4 that was hit hardest by the regulatory shocks. We estimate the following IV regression where BoA lending growth is instrumented with 2008 BoA share,¹⁵ and the control variables are the same as the baseline specification (3):

$$Growth_{county}^{BoA} = \theta Share_{county}^{08BoA} + \Delta Big4 Share_{county}^{2008} + Controls + \eta_{county}, \quad (A1)$$

$$Growth_{county}^{LenderClass} = \psi \left(\widehat{Growth_{county}^{BoA}} \right) + \Omega Big4 Share_{county}^{2008} + Controls + \epsilon_{county}. \quad (A2)$$

Table A.3 presents the results. The coefficient of interest is the point estimate on BoA lending growth (column 1), which is economically and statistically significant at -0.292 . A 10-pp lower BoA growth corresponds to a 3-pp higher small bank growth, even conditional on the ex ante Big4 share. The coefficient for the 2008 Big4 share itself is large and statistically significant, indicating that areas with greater Big4 share in 2008 experienced larger growth in small bank lending. The estimates for other lender classes (columns 2–6) are much smaller and often statistically insignificant. In sum, Table A.3 shows that even after accounting for the baseline heterogeneity in level of 2008 ex ante Big4 share, small bank lending grew significantly more in areas in which Big4 constituents were hit harder with regulatory shocks (as proxied by BoA exposure).

Appendix B. Reallocation of Lending within Small Banks

This appendix section presents a deeper investigation of how small banks reallocate their lending. We ask whether small bank substitution is happening (1) within-bank (individual small banks adjusting their lending footprint toward areas most affected by the shocks); (2) across banks (small banks in areas affected by the shock grow most and those in unaffected areas grow the least); or (3) a combination of both within-bank and across banks. We disaggregate data to the lender $\times$ county level for this analysis.

B.1 Within-Small-Bank Reallocation

Reallocation of lending within a lender can come from the lender entering new counties or changing the level of activities where they are already present. We first consider entry by defining *Entry* equal to one in a given county if the lender is present in 2015, but not 2008, while counties in which the lender was present in 2008 are coded zero. We test whether the exposure to the Big4 retreat of new counties that individual lenders tend to enter is different from the exposure to the Big4 retreat

¹⁵ Alternatively, we could instrument using the penalties assessed to BoA. Those sources of variation are at the bank level and are weighted by the county origination shares, so the identifying variation of such instruments would be the same as using the 2008 BoA share.

within their existing lending footprint. Similar to our main test, we use the initial conditions of the Big4 in a given county as an instrument for their subsequent growth.

$$Growth_{county}^{Big4} = \lambda Share_{county}^{08Big4} + \zeta_{stateFE} + \nu_{lender} + \eta_{county} \quad (B3)$$

$$Entry_{c,l} = \omega \left(\widehat{Growth_{county}^{Big4}} \right) + \xi_{stateFE} + \nu_{lender} + \epsilon_{county} \quad (B4)$$

The setup above includes individual lender fixed effects, which capture lender-specific drivers of changes in mortgage lending activities, including shocks to the bank's capital, regulatory pressure, or overall commitment to mortgage lending. The point estimate on Big4 growth indicates whether the new counties the lender enters have higher or lower levels of Big4 growth, relative to the degree of Big4 growth faced by the same lenders' prior footprint. Column 1 of Table A.4A presents the IV estimates and shows that small banks growth on the extensive margin is more likely to be in counties that experienced a larger Big4 retreat. The point estimate indicates that small banks are 1.5 pp more likely to enter a county with where Big4 growth is 10 pp lower. In column 2, we consider only counties in which the lender had a presence in 2008 and reestimate the test with the dependent variable equal to 100 if there was positive growth for the lender in the given county and zero otherwise. We find that a 10-pp lower Big4 growth within a given small bank's lending footprint corresponds to roughly 2-pp higher likelihood the small bank will have positive loan growth in that county from 2008 to 2015. While economically modest, these within-lender results provided evidence that individual banks were adjusting their lending according to the degree of Big4 withdrawal.

B.2 Across Small Bank Differences in Growth

To examine the presence of differences in loan growth across small banks according to their overall exposure to the Big4 retreat, we aggregate our data to the individual lender level. To measure a given lender's overall exposure to the Big4 retreat, we compute that particular lender's weighted average exposure to the Big4 retreat as follows:

$$WtGrowth_l^{Big4} = 100 * \frac{\sum_c Originations_{c,l,2008} * Growth_c^{Big4}}{\sum_c Originations_{c,l,2008}}. \quad (B5)$$

Column 3 presents the results from an IV regression of the lender-level percentage growth in their total lending from 2008 to 2015 on this weighted average exposure to Big4 lending growth. We use the lender-specific weighted average share of the Big4 (defined analogously to Equation (B5)) as the instrument. The results indicate that small banks exposed to a 10-pp lower (weighted average) growth by the Big4 corresponds to 3.4-pp higher lending growth. In column 4, the dependent variable is an indicator equal to 100 if the lenders growth was positive. We find that a 10-pp lower Big4 growth corresponds to an 8.8-pp higher likelihood of positive growth. In sum, we find evidence that our main results are driven by a combination of within-lender reallocation and across lender differential growth rates according to Big4 exposure. The size of the estimates in Table A.4 suggests that the across small bank differences play a larger role than within-small-bank reallocation. In panel B, we repeat the analysis using the exposure to g-fee shocks and arrive at similar conclusions.

Table A.1
County characteristics by Big4 growth and initial conditions

A. County characteristics by Big4 county-level growth, 2008–2015					
	Big4 growth quintile				
	1	2	3	4	5
Big4 growth, 2008–2015	–32.84 [–30.54]	–23.59 [–23.48]	–19.05 [–19.05]	–15.02 [–15.02]	–9.30 [–10.11]
Population	68.26 [19.27]	132.89 [38.69]	105.70 [34.61]	89.87 [31.17]	49.46 [21.69]
Minority	0.24 [0.15]	0.25 [0.20]	0.22 [0.16]	0.18 [0.10]	0.15 [0.08]
Income	32.99 [31.50]	34.12 [32.48]	34.69 [32.71]	33.78 [32.02]	32.67 [31.13]
Subprime	0.33 [0.32]	0.34 [0.33]	0.34 [0.33]	0.33 [0.33]	0.32 [0.31]
Age	39.73 [39.60]	38.94 [38.80]	39.10 [39.10]	39.31 [39.60]	39.87 [39.80]
HPgrowth	–1.69 [–2.60]	–3.86 [–4.98]	–2.51 [–3.51]	–1.24 [–1.96]	–1.38 [–2.27]
HHI	0.34 [0.28]	0.28 [0.23]	0.29 [0.24]	0.28 [0.24]	0.33 [0.27]
B. County characteristics by 2008 Big4 county-level origination share					
	Big4 share quintile				
	1	2	3	4	5
Big4 share 2008	13.44 [14.29]	20.25 [20.27]	25.80 [25.77]	32.07 [32.08]	44.12 [41.90]
Population	36.06 [21.69]	56.19 [26.41]	92.73 [32.79]	106.90 [36.06]	154.48 [24.89]
Minority	0.14 [0.07]	0.18 [0.10]	0.22 [0.16]	0.24 [0.19]	0.25 [0.19]
Income	31.09 [30.01]	32.00 [30.99]	33.76 [32.01]	34.29 [32.82]	37.12 [34.71]
Subprime	0.33 [0.32]	0.34 [0.34]	0.35 [0.34]	0.33 [0.33]	0.30 [0.28]
Age	39.68 [39.75]	39.60 [39.60]	39.04 [39.20]	38.71 [38.70]	39.91 [39.70]
HPgrowth	–1.78 [–2.10]	–1.86 [–2.36]	–2.23 [–3.21]	–2.66 [–4.42]	–2.15 [–3.53]
HHI	0.32 [0.27]	0.30 [0.25]	0.30 [0.24]	0.28 [0.23]	0.30 [0.24]

This table presents the mean (median) county characteristics for 2008 according to the amount county-level Big4 mortgage origination lending growth shares from 2008 to 2015 (panel A) and according to the 2008 Big4 share of originations (panel B). *Big4* represents Bank of America (*BoA*), Citi, JP Morgan, and Wells Fargo; *Population* is the 2008 population in thousands; *Minority* is the 2008 non-white share of the population; *Income* is the 2008 per capita income in thousands; *Subprime* is the 2008 share of the population with FICO below 660; *Age* is the median age in the population; *HHI* is the 2008 county banking competition base on deposits; and *HPgrowth* is the county-level house price growth over 2008–2015.

Table A.2

Using further lags of the Big4 share to instrument for the Big4 retreat

	2005 Big4 share as IV			2002 Big4 share as IV		
	(1) Small banks	(2) Shadow banks	(3) Fintech	(4) Small banks	(5) Shadow banks	(6) Fintech
$\widehat{Growth}_{county}^{Big4}$	-0.908*** (< 0.01)	0.006 (0.94)	-0.079** (0.05)	-0.799*** (< 0.01)	-0.021 (0.85)	-0.036 (0.36)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,982	2,982	2,982
R^2	.146	.146	.064	.155	.149	.067
F-stat	169	169	169	149	149	149

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents IV estimates from the regressions of county-level growth in mortgage originations for respective lender classes from 2008 to 2015 on the county-level growth in Big4 originations during this time period. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo, *Small banks* includes banks with assets less than \$10 billion, *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies), *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018), *Large banks* includes banks with assets between \$50 billion and \$1 trillion, and *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion. The dependent variables are the respective county-level loan growth from 2008 to 2015 for each lender class, and $\widehat{Growth}_{county}^{Big4}$ represents the loan growth for the Big4, which is instrumented in these regressions with the 2005 county-level share of the Big4 in columns 1–3, and the 2002 county-level share of the Big4 in columns 4–6. Control variables are the same as those in Table 4 and include state fixed effects. *F-stat* is the F-statistic from the first-stage regression. Standard errors are clustered by MSA.

Table A.3

Exploiting within-Big4 variation in regulatory shocks

	(1) Small banks	(2) Shadow banks	(3) Fintech	(4) Large banks	(5) Interm banks	(6) Credit unions
$\widehat{Growth}_{county}^{BoA}$	-0.292*** (0.01)	-0.129 (0.15)	-0.089*** (< 0.01)	-0.038 (0.27)	-0.080* (0.10)	-0.042 (0.21)
$\widehat{Big4\ share}_{county}^{2008}$	0.575*** (< 0.01)	0.200*** (< 0.01)	0.071*** (< 0.01)	0.046*** (0.01)	-0.010 (0.50)	0.006 (0.76)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,983	2,983	2,983
R^2	.144	.149	.091	.034	.023	.020

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents IV estimates from the regressions of county-level growth in mortgage originations for each lender class from 2008 to 2015 on the county-level growth in Bank of America originations during this time period ($\widehat{Growth}_{county}^{BoA}$). *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo, *Small banks* includes banks with assets less than \$10 billion, *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies), *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018), *Large banks* includes banks with assets between \$50 billion and \$1 trillion, and *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion. $\widehat{Growth}_{county}^{BoA}$ is instrumented with the 2008 county-level origination share of Bank of America. Control variables include 2008 county-level Big4 share of originations ($\widehat{Big4\ Share}_{county}^{2008}$), logarithm of the 2008 population, non-white share of the population, per capita income, median age, share of the population with FICO below 660, county banking HHI based on deposits, house price growth, the logarithm of the number of banks with a physical branch in the county, the logarithm of the number of banks with at least one mortgage loan in the county, and the logarithm of the number of shadow banks or fintech lenders with at least one mortgage loan in the county. All regressions include state fixed effects. The F-statistic for the first-stage regression is 12,591. Standard errors are clustered by MSA.

Table A.4
Small bank reallocation: Within and across banks

	Within bank		Across banks	
	(1) Entry	(2) (Growth>0)	(3) Growth	(4) (Growth>0)
<i>A. Big4 exposure and individual small bank growth</i>				
$Growth_{county}^{Big4}$	-0.150*** (< 0.01)	-0.196*** (< 0.01)		
$WtGrowth^{Big4}$			-0.335*** (< 0.01)	-0.876*** (< 0.01)
State FE	Yes	Yes	No	No
Lender FE	Yes	Yes	No	No
Observations	141,427	75,133	2,572	2,572
F-stat	1,150	1,099	1,497	1,497
<i>B. G-fee shocks and individual small bank growth</i>				
Top10GSELenderShare2011	0.051*** (0.01)	0.077*** (0.01)		
$WtTop10GSELenderShare$			0.322*** (< 0.01)	0.492*** (< 0.01)
State FE	Yes	Yes	No	No
Lender FE	Yes	Yes	No	No
Observations	138,453	78,119	2,850	2,850

Columns 1 and 2 of panel A present IV estimates from the regressions of growth in mortgage originations (entry or positive growth, both defined as indicators equal to 100 if true) for each small bank in each county for which they do business in either 2008 or 2018 class from 2008 to 2015 on the county-level growth in Big4 originations during this time period. The dependent variable in columns 3 is overall bank-level growth from 2008 to 2015, and column 4 is an indicator variable equal to 100 if that bank-level growth was positive. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; and $Growth_{county}^{Big4}$ represents the loan growth for the Big4 is instrumented with the 2008 county-level origination share of the Big4. $WtGrowth^{Big4}$ is the weighted average Big4 growth exposure for a given lender as defined in Equation (B5), which is instrumented in these regressions with the 2008 county-level weighted share of the Big4. The regressions in panel B are similarly defined for 2011–2015 growth only using Top10GSELenderShare2011 (as in Table 7) in the place of Big4 growth $Growth_{county}^{Big4}$. The *F-stat* is the first-stage F-statistic. Standard errors are clustered by lender and county in columns 1 and 2 of the panels and are heteroscedasticity robust in columns 3 and 4.

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